Natural Language Processing Project

Combined BERT and SBERT Embeddings for Temporal Story-Point Estimation

Problem Statement 10 · Within-project · Temporal ordering

GROUP MEMBERS	[ADD GROUP MEMBER NAMES AND IDs]
GITHUB	github.com/mhdh9632/tawos-bert-sbert-story-point-estimation
EXECUTED COLAB	Open completed experiment
DATE	11 November 2026

KEY RESULT

Combined BERT + SBERT achieved final-test MAE 3.9416 and MdAE 1.6988, the lowest errors among all evaluated representations.

Abstract

This project investigates whether combining contextual word-level information from BERT with sentence-level semantic information from SBERT improves software story-point estimation for later issues within the same project. TAWOS issues were filtered for usable text, dates, and story-point labels while excluding records changed after estimation. Lsstcorp Data Management (DM) supplied 14,782 eligible issues. The oldest 11,825 formed development data and the newest 2,957 formed the held-out test set. Five expanding-window validation folds preserved chronology. Frozen bert-base-uncased and all-MiniLM-L6-v2 encoders produced 768- and 384-dimensional vectors; concatenation produced 1,152 features. The combined model achieved final-test MAE 3.9416 and MdAE 1.6988, outperforming both baselines. Validation MdAE was mixed, so the result supports fusion for this target period without establishing universal superiority.

1. Introduction

Agile teams use story points to express relative issue effort or complexity. Estimation depends on technical detail, project vocabulary, and historical practice. NLP models can learn relationships between issue descriptions and previously assigned points, but evaluation must reflect deployment: past issues should predict future ones.

Problem Statement 10 asks whether combined word and sentence embeddings effectively predict story points of later issues within the same project. The research question is: **Does concatenating BERT-derived contextual representations and SBERT sentence embeddings reduce MAE and MdAE on a chronological held-out period?**

2. Dataset

2.1 TAWOS and modelling fields

TAWOS V1.1 contains 458,232 issues from 39 open-source projects in 12 public Jira repositories. The model used Issue_Key, Title, Description_Text, Creation_Date, Story_Point, Project_ID, and the three changed-after-estimation flags. Resolution, time spent, total effort, comments, and other post-outcome fields were excluded to prevent leakage.

2.2 Eligibility and project selection

Eligible issues required usable title or description text, a valid timestamp, a finite non-negative label, a recognised project, and no indication that title, description, or story point changed after estimation. Title and description were joined with a [SEP] marker. A memory-bounded parser extracted only required SQL columns and cached a compact CSV in Drive.

The notebook selected DM because it had the largest eligible sample: 14,782 issues dated 4 March 2014 to 22 October 2020. Median story points were 2.0 and the mean was 5.9843, indicating a right-skewed target distribution.

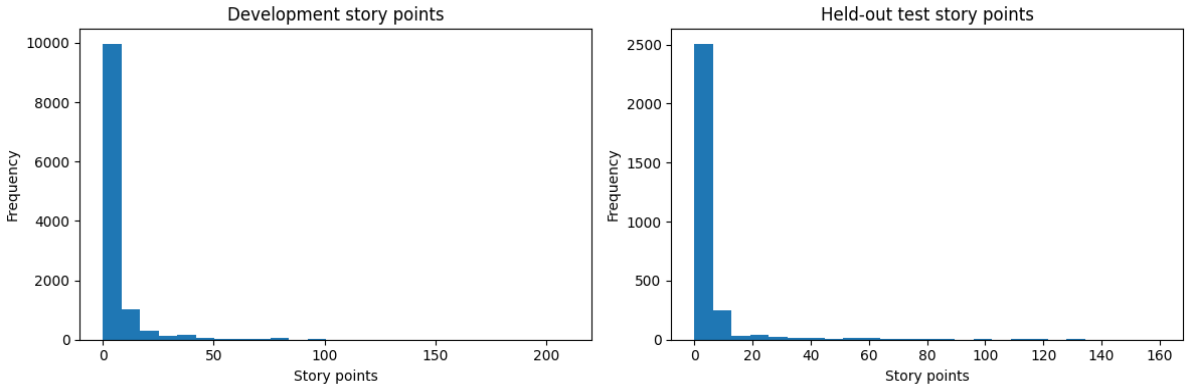


Figure 1. Story-point distributions in development and held-out test data.

3. Experimental Methodology

3.1 Temporal holdout

Issues were ordered by Creation_Date, with issue ID for stable tie ordering. The newest 20% was reserved before model selection. Equal timestamps were kept on the same side of a boundary. The final test set did not affect scaling, early stopping, or epoch selection.

Partition	Issues	Start	End
Development	11,825	2014-03-04 14:51	2019-11-07 22:44
Final test	2,957	2019-11-07 22:50	2020-10-22 14:51

3.2 Five expanding-window folds

Fold	Train	Validation	Train end	Validation period
1	1,952	1,955	2016-02-03	2016-02 to 2016-11
2	3,907	1,962	2016-11-30	2016-11 to 2017-09
3	5,869	1,993	2017-09-28	2017-09 to 2018-08
4	7,862	1,994	2018-08-06	2018-08 to 2019-04
5	9,856	1,969	2019-04-08	2019-04 to 2019-11

3.3 Representations and prediction model

- **BERT-only:** bert-base-uncased contextual token states, attention-mask-aware mean pooled to 768 dimensions.
- **SBERT-only:** all-MiniLM-L6-v2 semantic sentence vectors with 384 dimensions.
- **Combined:** concatenated BERT and SBERT vectors with 1,152 dimensions.
- **Regressor:** Linear(256) → ReLU → Dropout(0.20) → Linear(64) → ReLU → Dropout(0.10) → Linear(1).

Encoders were frozen. Features and targets were standardised from training data only. AdamW used learning rate 0.001, weight decay 0.0001, batch size 64, Smooth L1 loss, maximum 50 epochs, patience 7, and seed 42. Negative predictions were clipped to zero. Median best epoch across folds set final training duration.

4. Results

4.1 Five-fold validation summary

Representation	MAE mean	MAE SD	MdAE mean	MdAE SD
BERT	4.5545	0.2809	2.0266	0.0825
SBERT	4.5884	0.3763	2.0242	0.2154
Combined	4.5239	0.2999	2.0961	0.1090

Combined embeddings achieved the lowest mean validation MAE (4.5239), but SBERT achieved the lowest mean validation MdAE (2.0242). Fusion therefore did not uniformly dominate both baselines during model development.

4.2 Final held-out test

Representation	MAE	MdAE	Epochs	Dimensions
BERT	4.3400	2.1191	3	768
SBERT	4.4238	2.0220	1	384
Combined	3.9416	1.6988	3	1,152

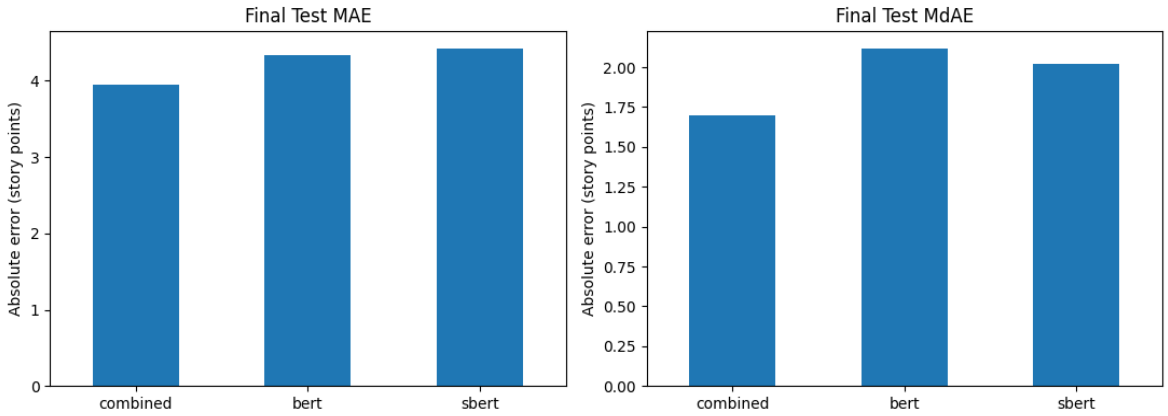


Figure 2. Final held-out MAE and MdAE; lower is better.

Combined embeddings achieved the lowest final error on both metrics. Against BERT, fusion reduced MAE by 0.3984 points (9.18%) and MdAE by 0.4203 (19.83%). Against SBERT, it reduced MAE by 0.4822 (10.90%) and MdAE by 0.3232 (15.98%).

5. Discussion

The held-out result indicates complementary information: BERT captures contextual technical wording while SBERT emphasises whole-issue meaning. Concatenation made both views available to the regressor and improved future-issue predictions. However, validation evidence was nuanced because combined MdAE was not best. The final result supports fusion for this target period, not universal superiority.

Combined final MAE (3.9416) exceeded MdAE (1.6988), indicating a subset of large errors. This is consistent with the right-skewed story-point distribution. Later folds generally had lower MAE, which may reflect larger training samples, temporal drift, or both.

6. Threats to Validity and Limitations

Internal validity

One seed was evaluated; broad hyperparameter search was not performed; representation dimensions differed even though heads and partitions were shared.

Construct validity

Story points are team-specific relative estimates; change-flag accuracy depends on TAWOS; descriptions beyond 256 tokens were truncated.

External validity

The result covers DM only; encoders were frozen; open-source Jira history may not represent industrial projects.

Conclusion validity

No repeated-seed confidence intervals or significance tests were conducted; one final temporal interval cannot describe all future periods.

7. Reproducibility

The repository includes clean and executed notebooks, exact configuration, split metadata, 15 fold records, summaries, plots, and final metrics. The 4.31 GB source dump and large embedding/model caches are excluded. Reproduction requires TAWOS.sql at MyDrive/TAWOS_NLP/data/raw/, a GPU-enabled Colab runtime, and running all cells in order. DEMO.md provides the presentation and verification sequence.

8. Conclusion

On 2,957 later DM issues, combined embeddings achieved MAE 3.9416 and MdAE 1.6988, outperforming both individual representations. This supports complementary contextual and sentence-level features for the selected project. Future work should repeat the experiment across projects and seeds, test software-domain encoders, compare fusion architectures, and evaluate statistical reliability.

References

1. Tawosi, V., Al-Subaihin, A., Moussa, R., and Sarro, F. A Versatile Dataset of Agile Open Source Software Projects. MSR 2022, 707–711. DOI: 10.1145/3524842.3528029.

2. Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL-HLT 2019.

3. Reimers, N. and Gurevych, I. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. EMNLP-IJCNLP 2019.

4. TAWOS V1.1 dataset. DOI: 10.5522/04/21308124.

Appendix A — Fold-level Validation Metrics

Representation	Fold	MAE	MdAE	Best epoch
BERT	1	4.6205	1.9664	3
BERT	2	4.7768	1.9281	4
BERT	3	4.7716	2.0285	3
BERT	4	4.5105	2.0769	3
BERT	5	4.0931	2.1333	8
SBERT	1	4.8302	2.3262	1
SBERT	2	4.8212	1.7208	1
SBERT	3	4.8659	2.0100	4
SBERT	4	4.4291	1.9999	1
SBERT	5	3.9957	2.0642	6
Combined	1	4.7524	2.0580	3
Combined	2	4.7501	2.0236	4
Combined	3	4.6822	2.1779	4
Combined	4	4.3648	1.9808	1
Combined	5	4.0699	2.2403	2

Appendix B — Recorded Configuration

Setting	Recorded value
Project	DM
Seed	42
Device	cuda
BERT model	bert-base-uncased
SBERT model	sentence-transformers/all-MiniLM-L6-v2
Maximum tokens	256
Fusion	Concatenation (768 + 384 = 1,152)
Training	AdamW; lr 0.001; wd 0.0001; batch 64
Selection	5 temporal folds; median best epoch
Metrics	MAE and MdAE